

SAT Unit 2.2

Quadratic function

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Overview

- 1 Different Forms of a Parabola
- 2 Properties of a Parabola
- 3 Quadratic Formula and X-Intercepts
- 4 Systems of Equations
- 5 Vieta's Theorem
- 6 Application
- 7 Practice

Parabolas

A parabola is a symmetric curve that satisfies a quadratic function. Below are different forms of its equation and key properties.

Different Forms of a Parabola

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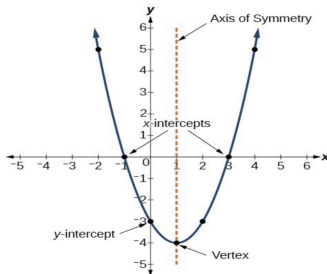
Standard Form

Standard Form:

$$f(x) = ax^2 + bx + c$$

where:

- a determines the direction and width of the parabola.
 - If $a > 0$, the parabola opens **upward** (concave up).
 - If $a < 0$, the parabola opens **downward** (concave down).
- b affects the location of the vertex along the x -axis.
- c is the y -intercept.

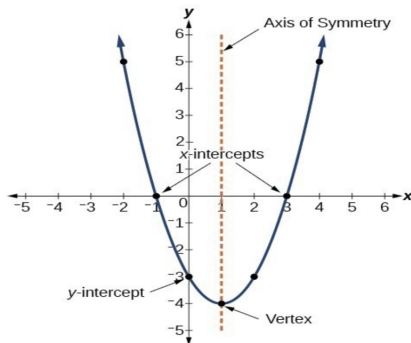


Vertex Form

Vertex Form:

$$f(x) = a(x - h)^2 + k$$

where (h, k) is the vertex of the parabola.

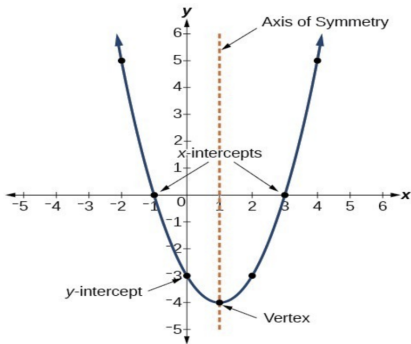


Factored Form

Factored Form:

$$f(x) = a(x - m)(x - n)$$

where m and n are the x -intercepts (roots) of the parabola.



Question: Quadratic Function from Three Conditions

Problem:

A quadratic function $f(x)$ satisfies the following:

- The vertex of the parabola is at $(3, -16)$.
- The graph passes through the point $(5, -12)$.
- One of the roots of $f(x)$ is 1.

Tasks:

- 1 Write the function in vertex form.
- 2 Convert your function to standard form.
- 3 Express the function in factored form.

Solution: Challenge Problem

Step 1: Vertex Form

$$f(x) = a(x - 3)^2 - 16$$

Plug in $(5, -12)$ to solve for a :

$$-12 = a(5 - 3)^2 - 16 \Rightarrow -12 = 4a - 16 \Rightarrow 4a = 4 \Rightarrow a = 1$$

$$\Rightarrow f(x) = (x - 3)^2 - 16$$

Step 2: Standard Form

$$f(x) = x^2 - 6x + 9 - 16 = x^2 - 6x - 7$$

Step 3: Factored Form

$$f(x) = (x - 7)(x + 1)$$

(One root is 1, so the other must be 7 since sum of roots = 6)



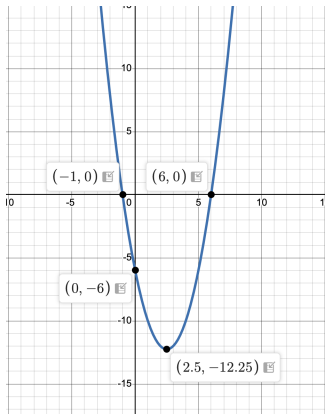
Vertex

Standard Form:

$$f(x) = ax^2 + bx + c$$

Vertex:

$$x = \frac{-b}{2a}, \quad y = f\left(\frac{-b}{2a}\right) = \frac{4ac - b^2}{4a}$$



Question: Minimum Value of Transformed Quadratic Function

The function $f(x) = 4x^2 + 64x + 262$. The function g is defined by:

$$g(x) = f(x + 5)$$

For what value of x does $g(x)$ reach its minimum?

Options:

A -13

B -8

C -5

D -3

Answer Explanation: Minimum of

$$g(x) = f(x + 5)$$

Correct Answer: -13

Step 1: Find the vertex of the quadratic function Since $f(x) = ax^2 + bx + c$, the vertex is at:

$$x = -\frac{b}{2a} = -\frac{64}{2(4)} = -8$$

So, the minimum of $f(x)$ occurs at $x = -8$

Step 2: Apply the transformation to $g(x) = f(x + 5)$ This is a horizontal shift **left by 5 units**. So the minimum of $g(x)$ occurs at:

$$x + 5 = -8 \Rightarrow x = -13$$

Final Answer: -13

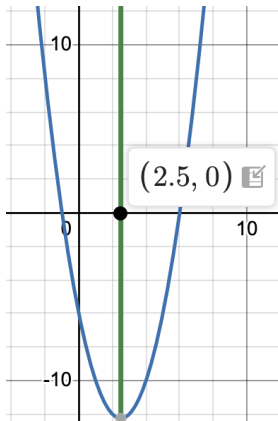
Axis of Symmetry

Standard Form:

$$f(x) = ax^2 + bx + c$$

Axis of Symmetry:

$$x = \frac{-b}{2a}$$



Question: Minimum of Quadratic Function (Factored Form)

The function is defined as:

$$f(x) = (x - 14)(x + 19)$$

For what value of x does $f(x)$ reach its minimum?

Options:

A -266

B -19

C $-\frac{33}{2}$

D $-\frac{5}{2}$

Answer Explanation: Minimum of

$$f(x) = (x - 14)(x + 19)$$

Correct Answer: D

Step 1: Use the axis of symmetry formula for a quadratic in factored form:

$$x = \frac{r_1 + r_2}{2}$$

where $r_1 = 14$, $r_2 = -19$

$$x = \frac{14 + (-19)}{2} = \frac{-5}{2} = -\frac{5}{2}$$

Question: Properties of a Quadratic Function

The function is given as:

$$f(x) = ax^2 + 4x + c$$

where a and c are constants. The parabola opens upward and it is also given that:

- $f(-9) = f(3)$

What is the value of a ?

Answer Explanation: Use of Symmetry and Vertex

We are told:

$$f(x) = ax^2 + 4x + c \quad \text{and} \quad f(-9) = f(3)$$

Step 1: Use symmetry of a parabola. If $f(-9) = f(3)$, then these points are symmetric about the vertex.

So, the axis of symmetry (i.e. x-value of vertex) is:

$$x = \frac{-9 + 3}{2} = -3$$

Step 2: Use vertex formula to confirm. The x-coordinate of the vertex of a quadratic is given by:

$$x = \frac{-b}{2a}$$

Here, $f(x) = ax^2 + 4x + c$, so:

$$\frac{-4}{2a} = -3 \Rightarrow a = \frac{2}{3}$$

Question

In the xy -plane, a parabola has vertex $\left(\frac{7}{3}, -4\right)$ and intersects the x -axis at $(10, 0)$ and $(a, 0)$, where a is a constant.

What is the value of a ?

A) -10

B) $-\frac{23}{3}$

C) $-\frac{19}{3}$

D) $-\frac{16}{3}$

Answer

We are given the vertex $\left(\frac{7}{3}, -4\right)$ and the x-intercepts $(10, 0)$ and $(a, 0)$.

So the parabola has roots at $x = 10$ and $x = a$, and the vertex is at the midpoint of the roots.

Use midpoint formula:

$$\frac{10 + a}{2} = \frac{7}{3} \Rightarrow 10 + a = \frac{14}{3} \Rightarrow a = \frac{14}{3} - 10 = \frac{14 - 30}{3} = -\frac{16}{3}$$

Correct answer: D

Question

The function

$$f(x) = (x - 3)(x - k)$$

is a quadratic, where k are constants.

The graph of $y = f(x)$ in the xy -plane is a parabola with vertex $(5, -32)$.

What is the value of k ?

- A) 2
- B) 5
- C) 7
- D) 8

Answer

C



Discriminant of a Quadratic Equation

General Form of a Quadratic:

$$ax^2 + bx + c = 0$$

Discriminant Formula:

$$\Delta = b^2 - 4ac$$

Why is the discriminant important?

- Tells us how many and what type of solutions the quadratic equation has.

Cases:

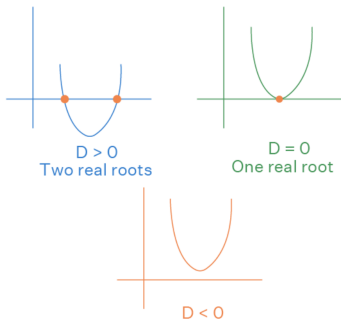
- $\Delta > 0$: Two distinct real solutions
- $\Delta = 0$: One real solution (a repeated root)
- $\Delta < 0$: No real solutions (complex roots)

Graphical Meaning of the Discriminant

How the Discriminant relates to the graph of $y = ax^2 + bx + c$:

- $\Delta > 0$: The parabola intersects the x-axis at **two points**.
- $\Delta = 0$: The parabola is **tangent** to the x-axis (one point of contact).
- $\Delta < 0$: The parabola does **not intersect** the x-axis.

Implication: You can *predict* how many real solutions an equation has by simply analyzing its discriminant.



Question

$$2x^2 - 13x + 21 = 0$$

How many distinct real solutions does the given equation have?

- A) Zero
- B) Exactly one
- C) Exactly two
- D) Infinitely many

Answer

We are given the quadratic equation:

$$2x^2 - 13x + 21 = 0$$

Use the discriminant to determine the number of real solutions:

$$\Delta = b^2 - 4ac = (-13)^2 - 4(2)(21) = 169 - 168 = 1$$

Since $\Delta > 0$, the equation has two distinct real solutions.

Correct answer: C

Question: Discriminant and No Real Solutions

Consider the quadratic equation:

$$x^2 - 34x + c = 0$$

In the given equation, c is a constant. The equation has **no real solutions** if $c > n$. What is the least possible value of n ?

Answer Explanation: Using the Discriminant

A quadratic equation $ax^2 + bx + c = 0$ has:

- **Two real solutions** if $b^2 - 4ac > 0$
- **One real solution** if $b^2 - 4ac = 0$
- **No real solution** if $b^2 - 4ac < 0$

In our equation:

$$x^2 - 34x + c = 0 \Rightarrow a = 1, \quad b = -34, \quad c = c$$

Compute the discriminant:

$$\Delta = (-34)^2 - 4(1)(c) = 1156 - 4c$$

Set discriminant to be less than 0:

$$1156 - 4c < 0 \Rightarrow 4c > 1156 \Rightarrow c > 289$$

Therefore, the equation has no real solutions when:

$$c > 289 \Rightarrow \boxed{n = 289}$$

The Quadratic Formula

To solve any quadratic equation:

$$ax^2 + bx + c = 0$$

Use the Quadratic Formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Steps to Use the Formula:

- 1 Identify a , b , and c from the equation.
- 2 Plug the values into the formula.
- 3 Simplify the square root (use the discriminant).
- 4 Solve for both values using $+$ and $-$.

Note: Always check the discriminant before solving to understand the type of roots.

Question: Quadratic Equation with Radical Solutions

Consider the equation:

$$2x^2 - 2 = 2x + 3$$

Which of the following is a solution to the equation above?

Options:

A 2

B $1 - \sqrt{11}$

C $\frac{1}{2} + \sqrt{11}$

D $\frac{1 + \sqrt{11}}{2}$

Answer Explanation: Quadratic Equation with Radical Solutions

Correct Answer: D: $\frac{1+\sqrt{11}}{2}$

Solution:

- Start by rearranging the equation:

$$2x^2 - 2 = 2x + 3 \Rightarrow 2x^2 - 2x - 5 = 0$$

- Use the quadratic formula:

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(2)(-5)}}{2(2)} = \frac{2 \pm \sqrt{4 + 40}}{4} = \frac{2 \pm \sqrt{44}}{4}$$

- Therefore, the solutions are:

$$x = \frac{1 + \sqrt{11}}{2}, \quad x = \frac{1 - \sqrt{11}}{2}$$

Final Answer: $\boxed{\frac{1 + \sqrt{11}}{2}}$

Question: Solving a Quadratic Equation Involving Radicals

Solve the quadratic equation:

$$x^2 - 2x - 9 = 0$$

One solution to the equation can be written in the form $1 + \sqrt{k}$, where k is a constant.

What is the value of k ?

Options:

- A 8
- B 10
- C 20
- D 40

Answer Explanation: Solving a Quadratic Equation Involving Radicals

Correct Answer: C: 20

Solution:

- Use the quadratic formula:

$$x = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(-9)}}{2(1)} = \frac{2 \pm \sqrt{4 + 36}}{2} = \frac{2 \pm \sqrt{40}}{2}$$

- Simplify:

$$x = \frac{2 \pm 2\sqrt{10}}{2} = 1 \pm \sqrt{10}$$

- One solution is $1 + \sqrt{10}$, so we identify:

$$\sqrt{k} = \sqrt{10} \Rightarrow k = 10$$

Final Answer: 10

Question: Smallest Solution to a Radical Equation

Solve the equation:

$$\sqrt{(x - 2)^2} = \sqrt{3x + 34}$$

What is the **smallest solution** to the given equation?

Answer Explanation: Solving Radical Equation

Square both sides:

$$(x-2)^2 = 3x+34 \Rightarrow x^2-4x+4 = 3x+34 \Rightarrow x^2-7x-30 = 0 \Rightarrow (x-10)(x+3)$$

Now check for extraneous solutions by plugging into the original equation:

- For $x = 10$:

$$\sqrt{(10-2)^2} = \sqrt{3(10)+34} \Rightarrow \sqrt{64} = \sqrt{64} \quad \text{valid}$$

- For $x = -3$:

$$\sqrt{(-3-2)^2} = \sqrt{3(-3)+34} \Rightarrow \sqrt{25} = \sqrt{25} \quad \text{valid}$$

Case 2: $-(x-2) = \sqrt{3x+34}$

This will yield the same roots already considered since squaring removes signs.

Final Answer: -3

Question: Solving a Radical Equation

Solve the equation:

$$\sqrt{2x + 6} + 4 = x + 3$$

What is the solution set?

Options:

- A. $\{-1\}$
- B. $\{5\}$
- C. $\{-1, 5\}$
- D. $\{0, -1, 5\}$

Answer Explanation: Solving Radical Equation

We start with the given equation:

$$\sqrt{2x+6} + 4 = x + 3$$

Subtract 4 from both sides:

$$\sqrt{2x+6} = x - 1$$

Now square both sides:

$$2x + 6 = (x - 1)^2 \Rightarrow 2x + 6 = x^2 - 2x + 1$$

Move all terms to one side:

$$0 = x^2 - 4x - 5 \Rightarrow (x - 5)(x + 1) = 0 \Rightarrow x = 5 \quad \text{or} \quad x = -1$$

Answer Explanation: Solving Radical Equation

Check for extraneous solutions:

For $x = 5$:

$$\sqrt{2(5) + 6} + 4 = \sqrt{16} + 4 = 4 + 4 = 8 \quad \text{and} \quad x + 3 = 8 \quad \Rightarrow \quad \text{valid}$$

For $x = -1$:

$$\sqrt{2(-1) + 6} + 4 = \sqrt{4} + 4 = 2 + 4 = 6 \quad \text{and} \quad -1 + 3 = 2 \quad \Rightarrow \quad \text{not valid}$$

Final Answer: $\{5\}$

What is a System of Equations?

- A **system of equations** is a set of two or more equations with the same variables.
- The solution is the point(s) that make all equations true at the same time.

Example:

$$\begin{cases} y = 2x + 1 \\ y = -x + 4 \end{cases}$$

Find the value of x and y that satisfies both equations.

Solving by Graphing

- Plot both equations into the Desmos.
- The intersection point is the solution to the system.

$$\begin{cases} y = x^2 \\ y = 2x + 3 \end{cases}$$

Graph to find the point(s) of intersection.

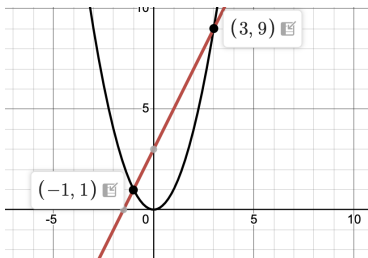
Solving by Graphing

- Plot both equations into the Desmos.
- The intersection point is the solution to the system.

$$\begin{cases} y = x^2 \\ y = 2x + 3 \end{cases}$$

Graph to find the point(s) of intersection.

Solution: The two curves intersect at points where $x^2 = 2x + 3$.



Exactly One Real Solution

Given the system:

$$\begin{cases} y = -1.5 \\ y = x^2 + 8x + a \end{cases}$$

Question: If the system has **exactly one real solution**, what is the value of a ?

Exactly One Real Solution

Given the system:

$$\begin{cases} y = -1.5 \\ y = x^2 + 8x + a \end{cases}$$

Question: If the system has **exactly one real solution**, what is the value of a ?

Step 1: Set the two equations equal to each other:

$$x^2 + 8x + a = -1.5 \Rightarrow x^2 + 8x + (a + 1.5) = 0$$

Step 2: For a quadratic to have exactly one real solution, the discriminant must be zero:

$$b^2 - 4ac = 0 \Rightarrow 8^2 - 4(1)(a + 1.5) = 0$$

Step 3: Solve:

$$64 - 4(a + 1.5) = 0 \Rightarrow a + 1.5 = 16 \Rightarrow a = 14.5$$

Vieta's Theorem for Quadratic Equations

Given: A quadratic equation in standard form

$$ax^2 + bx + c = 0$$

with roots r_1 and r_2 .

Vieta's Theorem:

- Sum of the roots: $r_1 + r_2 = -\frac{b}{a}$
- Product of the roots: $r_1 \cdot r_2 = \frac{c}{a}$

Why is this useful?

- Solve problems involving the relationships between roots without solving the equation.
- Helps in constructing quadratics from given roots.

Question: Sum of the Solutions of a Quadratic Equation

What is the sum of the solutions to the given equation?

$$x^2 - 40x - 10 = 0$$

Options:

- A. 0
- B. 5
- C. 10
- D. 40

Answer Explanation: Sum of the Solutions

We are given a quadratic equation:

$$x^2 - 40x - 10 = 0$$

For any quadratic equation of the form:

$$ax^2 + bx + c = 0,$$

the sum of the solutions is given by the formula:

$$\text{Sum} = -\frac{b}{a}.$$

Here, $a = 1$, $b = -40$. Thus,

$$\text{Sum} = -\frac{-40}{1} = 40.$$

Final Answer: 40

Question: Quadratic Product of Roots and Constants

In the given equation:

$$57x^2 + (57b + a)x + ab = 0,$$

where a and b are positive constants, the product of the solutions is given by kab , where k is a constant.

What is the value of k ?

Options:

- A. $\frac{1}{57}$
- B. $\frac{1}{19}$
- C. 1
- D. 57

Answer Explanation: Product of Solutions

We are given the quadratic:

$$57x^2 + (57b + a)x + ab = 0.$$

Using the quadratic formula properties, the product of the roots of

$$Ax^2 + Bx + C = 0$$

is:

$$\frac{C}{A}.$$

So here:

$$\text{Product of roots} = \frac{ab}{57}.$$

We are told this equals kab , so:

$$kab = \frac{ab}{57}.$$

Final Answer: $\boxed{\frac{1}{57}}$



Application of Quadratic Functions

Real-World Example: Projectile Motion

The height h (in meters) of a ball thrown into the air after t seconds is modeled by a quadratic function:

$$h(t) = -4.9t^2 + vt + h_0$$

where:

- -4.9 is the acceleration due to gravity (in m/s^2),
- v is the initial velocity (in m/s),
- h_0 is the initial height (in meters).

Key Features:

- The **maximum height** occurs at the vertex.
- The **time to reach the ground** is when $h(t) = 0$ (solve the equation).

Application: Use quadratic equations to model and solve problems involving motion, area, and economics.

Question: Understanding Intercepts in a Revenue Function

The revenue $f(x)$, in dollars, that a company receives from sales of a product is given by the function:

$$f(x) = -500x^2 + 25,000x,$$

where x is the unit price, in dollars, of the product.

The graph of $y = f(x)$ in the xy -plane intersects the x -axis at 0 and a .

What does a represent?

Options:

- A The revenue, in dollars, when the unit price of the product is \$0
- B The unit price, in dollars, of the product that will result in maximum revenue
- C The unit price, in dollars, of the product that will result in a revenue of \$0
- D The maximum revenue, in dollars, that the company can make

Answer Explanation: Understanding Intercepts in a Revenue Function

Correct Answer: C: The unit price, in dollars, of the product that will result in a revenue of \$0

Explanation:

- The function is a downward-opening parabola ($a = -500 < 0$).
- The x -intercepts of the graph represent values of x for which $f(x) = 0$, meaning revenue is \$0.
- At $x = 0$, revenue is 0 — this makes sense because selling at \$0 means no money is made.
- At $x = a$, the company again makes \$0 revenue — likely because the price is too high and no one buys it.
- Thus, a represents the price at which the revenue becomes zero again.

Final Answer: C

Question: Interpreting the Vertex of a Quadratic Function in Context

The function

$$f(x) = \frac{1}{9}(x - 7)^2 + 3$$

gives a metal ball's height above the ground $f(x)$, in inches, x seconds after it started moving on a track, where $0 \leq x \leq 10$.

Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

Options:

- A The metal ball's minimum height was 3 inches above the ground.
- B The metal ball's minimum height was 7 inches above the ground.
- C The metal ball's height was 3 inches above the ground when it started moving.
- D The metal ball's height was 7 inches above the ground when it started moving.

Answer Explanation: Vertex of a Quadratic in Real-World Context

Correct Answer: A: The metal ball's minimum height was 3 inches above the ground.

Explanation:

- The function is written in vertex form: $f(x) = a(x - h)^2 + k$, where the vertex is at (h, k) .
- Here, the vertex is at $(7, 3)$.
- Since the coefficient of the squared term is positive $(\frac{1}{9})$, the parabola opens upward, and the vertex represents a minimum.
- So, the ball reaches its **minimum height** of 3 inches **7 seconds** after it starts moving.

Final Answer: A



Question

The function $f(x) = 3x^2 + bx + c$ is a quadratic function, where b and c are constants.

The graph of $y = f(x)$ in the xy -plane does not intersect the x -axis. If $f(-7) = f(9)$, which of the following could be the value of $b + c$?

- A) -1
- B) -3
- C) -5
- D) -7

Answer I

We are given:

$$f(x) = 3x^2 + bx + c, \quad f(-7) = f(9)$$

Use symmetry: $f(-7) = f(9)$ implies that the axis of symmetry is the midpoint between -7 and 9 :

$$x = \frac{-7 + 9}{2} = 1 \Rightarrow \text{Vertex is at } x = 1$$

The vertex of a quadratic $f(x) = ax^2 + bx + c$ occurs at:

$$x = -\frac{b}{2a} = -\frac{b}{6}$$

Set equal to 1:

$$-\frac{b}{6} = 1 \Rightarrow b = -6$$

Now plug $b = -6$ into the original function:

$$f(x) = 3x^2 - 6x + c$$

Answer II

Now compute $b + c = -6 + c$, and check answer choices.

Try: - A) $-1 \Rightarrow c = 5$ - B) $-3 \Rightarrow c = 3$ - C) $-5 \Rightarrow c = 1$ - D) $-7 \Rightarrow c = -1$

We also know the graph does not intersect the x-axis $\rightarrow$ discriminant < 0 :

$$\Delta = b^2 - 4ac = (-6)^2 - 4(3)(c) = 36 - 12c$$

We want $36 - 12c < 0 \Rightarrow c > 3$

Only option that gives $c > 3$ is:

$A) -1 \Rightarrow b + c = -6 + 5 = -1$

Correct answer: A

Question: Properties of a Quadratic Function

The function is given as:

$$f(x) = ax^2 + 4x + c$$

where a and c are constants. The parabola opens upward and has a vertex at point (h, k) . It is also given that:

- $k < 0$
- $f(-9) = f(3)$

Which of the following must be true?

- I. $c < 0$
- II. $a \geq 1$

Options:

- A I only
- B II only
- C I and II
- D Neither I nor II

Answer Explanation: Use of Symmetry and Vertex

We are told:

$$f(x) = ax^2 + 4x + c \quad \text{and} \quad f(-9) = f(3)$$

Step 1: Use symmetry of a parabola. If $f(-9) = f(3)$, then these points are symmetric about the vertex.

So, the axis of symmetry (i.e. x-value of vertex) is:

$$x = \frac{-9 + 3}{2} = -3$$

Step 2: Use vertex formula to confirm. The x-coordinate of the vertex of a quadratic is given by:

$$x = \frac{-b}{2a}$$

Here, $f(x) = ax^2 + 4x + c$, so:

$$\frac{-4}{2a} = -3 \Rightarrow a = \frac{2}{3}$$

Answer Explanation: Use of Symmetry and Vertex

Thus, statement II: $a \geq 1$ is **not necessarily true**.

Step 3: What about $c < 0$? From the question, we know that $k < 0$. Therefore, by using **Vertex Formula**

$$\frac{4ac - b^2}{4a} < 0$$

, we can get that $c < 6$. Hence, Option II is wrong

Final Answer: D. Neither I nor II

Question: Parabola Vertex and Coefficient Sum

In the xy -plane, a parabola has vertex $(9, -14)$ and intersects the x -axis at two points. If the equation of the parabola is written in the form:

$$y = ax^2 + bx + c,$$

where a , b , and c are constants, which of the following could be the value of $a + b + c$?

Options:

A -23

B -19

C -14

D -12

Answer Explanation: Convert from Vertex to Standard Form

We are given:

- Vertex: $(9, -14)$
- Parabola intersects the x -axis at two points $\Rightarrow$ it opens upwards or downwards (not a single point).

Step 1: Use vertex form.

$$y = a(x - 9)^2 - 14$$

Step 2: Expand to standard form.

$$y = a(x^2 - 18x + 81) - 14 = ax^2 - 18ax + 81a - 14$$

So we identify:

$$a = a, \quad b = -18a, \quad c = 81a - 14$$

Step 3: Compute the sum $a + b + c$:

$$a + (-18a) + (81a - 14) = (a - 18a + 81a) - 14 = 64a - 14$$

Question: System of Equations - One Point of Intersection

Given the system of equations:

$$y = 2x^2 - 21x + 64$$

$$y = 3x + a$$

In the system, a is a constant. The graphs intersect at exactly one point (x, y) in the xy -plane. What is the value of x ?

(Can you do this question without using Quadratic Formula ?)

Options:

- A. -8
- B. -6
- C. 6
- D. 8

Answer I

We set the two expressions for y equal:

$$2x^2 - 21x + 64 = 3x + a$$

Move all terms to one side:

$$2x^2 - 24x + (64 - a) = 0$$

Since the graphs intersect at exactly one point, this quadratic must have exactly one solution. That means the discriminant is zero:

$$\Delta = b^2 - 4ac = 0$$

Here, $a = 2$, $b = -24$, $c = 64 - a$. So:

$$(-24)^2 - 4(2)(64 - a) = 0$$

$$576 - 8(64 - a) = 0 \Rightarrow 576 - 512 + 8a = 0 \Rightarrow 64 + 8a = 0 \Rightarrow a = -8$$

Now substitute $a = -8$ into the original system:

$$y = 3x - 8 \quad \text{and} \quad y = 2x^2 - 21x + 64$$

Answer II

Set equal again:

$$2x^2 - 21x + 64 = 3x - 8 \Rightarrow 2x^2 - 24x + 72 = 0$$

Divide by 2:

$$x^2 - 12x + 36 = 0 \Rightarrow (x - 6)^2 = 0 \Rightarrow \boxed{x = 6}$$

Final Answer: C. 6

Question

The function

$$f(x) = a(x - 3)(x - k)$$

is a quadratic, where a and k are constants.

The graph of $y = f(x)$ in the xy -plane is a parabola with vertex $(5, -32)$.

What is the value of a ?

- A) 2
- B) 5
- C) 6
- D) 8

Answer

Given: $f(x) = a(x - 3)(x - k)$ and vertex is at $(5, -32)$.

Step 1: Use symmetry — the x -coordinate of the vertex is the midpoint of the roots:

$$\frac{3 + k}{2} = 5 \Rightarrow 3 + k = 10 \Rightarrow k = 7$$

So the function is:

$$f(x) = a(x - 3)(x - 7)$$

Step 2: Plug in the vertex point $(5, -32)$:

$$f(5) = a(5 - 3)(5 - 7) = a(2)(-2) = -4a$$

$$-4a = -32 \Rightarrow a = 8$$

Correct answer: D

Thank You!

We appreciate your time and focus.

Novel Prep